Discrete Morse Theory

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September 25, 2025

1 Simplicial complexes

1.1 Affine independence

1.2 Geometric simplicial complex

1.3 Abstract simplicial complex

1.4 Equivalence

1.5 Simplicial maps

2 Collapsing

3 Discrete Morse functions

For K an abstract simplicial complex, a map $f : K \to \mathbb{R}$ is called a *discrete Morse function* if $\forall\ \sigma^k \in K$

(i) $e_1 = \left|\left\{\tau^{k-1} \in K \mid f(\tau) \geq f(\sigma)\right\}\right| \leq 1$;

(ii) $e_2 = \left|\left\{\tau^{k+1} \in K \mid f(\tau) \leq f(\sigma)\right\}\right| \leq 1$.

Every function g that *respects dimension* in the sense that for each $\sigma^{k-1} \subset \tau^k \in K$, $g(\sigma) < g(\tau)$ is a DMF. On the other hand, a DMF *almost* respects dimension by ensuring that each simplex can have one "exceptional" facet and one "exceptional" co-face; and in fact, not both. To arrive at a contradiction by assuming that for a face $\sigma \cup \left\{v_1\right\} \in K$, where $\sigma \in K$ and $v_1, v_2 \in K_0$, $f(\sigma) \geq f\left(\sigma \cup \left\{v_1\right\}\right) \geq f\left(\sigma \cup \left\{v_1, v_2\right\}\right)$, we have for $\sigma \cup \left\{v_2\right\} \in K$, $f\left(\sigma \cup \left\{v_1, v_2\right\}\right) > f\left(\sigma \cup \left\{v_2\right\}\right) > f(\sigma)$, the first strict inequality since the exceptional coface of σ is $\sigma \cup \left\{v_1\right\}$ and the second one since the exceptional face of $\sigma \cup \left\{v_1, v_2\right\}$ is $\sigma \cup \left\{v_1\right\}$. Simplices with exceptions form disjoint pairs of a simplex τ and its facet σ, termed *regular pairs*. It encodes an arrow $\sigma \to \tau$ in the sense that σ can be collapsed onto τ. Consecutive collapsings end up in a simplex which can't be collapsed any more, or a simplex without any exception, called a *critical simplex*.

If n_i is the number of critical simplices of dimension i in K, $\chi = \sum_{i=0}^{\infty} (-1)^i\, n_i$.

A *discrete vector field* is a disjoint collection of such pairs $\left(\sigma_i, \tau_i\right)$ where for each i, σ_i is a facet of τ_i. Each pair represents as an arrow $\to_i$. The disjointness means that each simplex can be a member of at most one pair in a vector field.

If a discrete vector field $\left\{\left(\sigma_i, \tau_i\right)\right\}_{i \in I}$ is given on a simplicial complex K, and $p \in \mathbb{N}$, a *p-path* is a sequence

$$\sigma_{i_1}^{p-1} \to \tau_{i_1}^{p} \geq \sigma_{i_2}^{p-1} \to \tau_{i_2}^{p} \geq \sigma_{i_3}^{p-1} \ldots \to \tau_{i_k}^{p} \geq \sigma_{i_{k+1}}^{p-1},$$

each arrow in the discrete vector field, and σ_{i_j} is a facet of $\tau_{i_{j-1}}$. It's a *cycle* is $\sigma_1 = \sigma_{k+1},\ k \geq 1$, and *acyclic* if it admits no cycle. A critical simplex can only appear as the last simplex in any p-path in a discrete vector field.

Regular pairs of a DMF form a discrete vector field. A discrete vector field formed by the regular pairs of a DMF is called a *gradient vector field*.

Theorem 1. *Each gradient vector field is acyclic. On the other hand, each acyclic discrete vector field on a simplicial complex K is induced by some DMF, i.e., is a gradient vector field.*

Proof. Given a DMF, in the induced discrete vector field, function values decrease along any p-path, since $f\left(\sigma_{i_j}\right) \geq f\left(\tau_{i_j}\right)$ being the only exception for $f\tau_{i_j}$, $f\left(\tau_{i_j}\right) > f\left(\sigma_{i_{j+1}}\right)$ for all j. Thus, *each gradient vector field is acyclic*. COMPLETE ☐

However, different DMF's on a simplicial complex might induce the same DVF, e.g., if f is a DMF, then so are e^f and $3f$, and all three induce the same DVF. Our primary interest in DVF's lies in their encodings of collapses and deformations-simplifications of a simplicial complex. A DMF represents a convenient but not unique way of encoding a DVF.

Proposition 2. *Suppose the critical simplices of an acyclic DVF on K form a subcomplex $L \leq K$. Then $\exists$ a collapse $K \to L$, and so, $K \simeq L$.*

For $\sigma \in K \setminus L$, we claim the existence of a regular pair (σ, τ) where σ is a *free face* of τ, whence we can perform an elementary collapse, and proceed using the same claim on the resulting complex. To prove the claim, let n denote the maximal dimension of a simplex in $K \setminus L$. There exist n-path(s) in the DVF (of which we take the maximal), and $\sigma \to \tau$ be its first regular pair. Now, $\sigma \leq \tau$, and if σ were a facet of some other simplex τ' in $K \setminus L$, then we can extend the n-path as $\tau' \geq \sigma \to \tau \geq \dots$, contradicting its maximality. On the other hand, if σ were a facet of some simplex τ' in L, then $\sigma \in L$ as L is a subcomplex, a contradiction. ☐

4 Morse homology

5 Computations

5.1 An easy simplex

5.2 Torus

6 Grid graphs

7 Computation with grid graph